

Participation and Spending in the Medicare Shared Savings Program

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Dissertation Defense

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Motivating Facts

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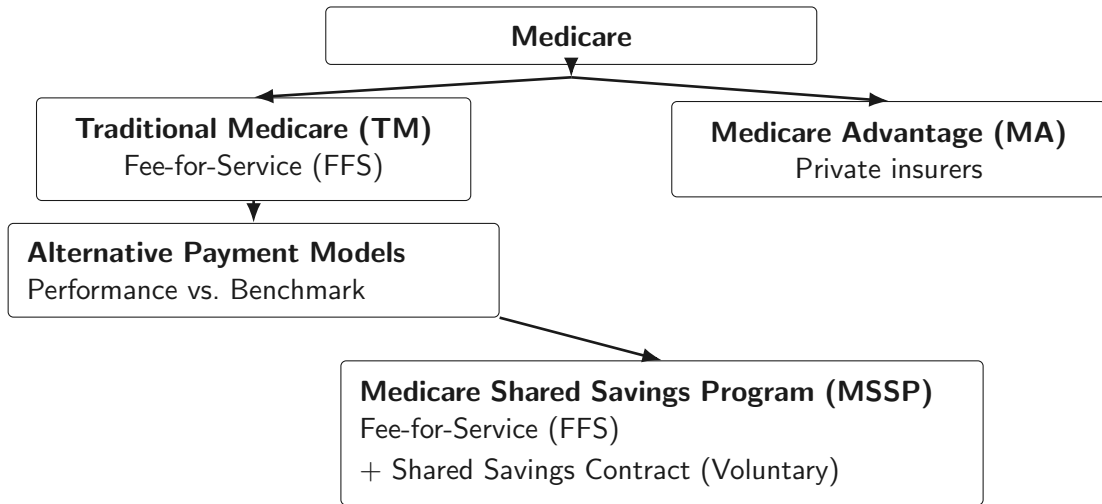
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- **Alternative Payment Models:** government initiatives to reduce overspending under Medicare.
- **The Medicare Shared Savings Program (MSSP)**
 - ▶ Aims to reduce spending compared to Medicare's FFS expenditure.
 - ▶ Makes it profitable for providers to **reduce the volume of unnecessary care.**

Institutional Setting



Medicare Shared Savings Program

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$$B_{it+1} = Y_{it}^H + \underbrace{\lambda(Y_{it}^R - Y_{it}^H)}_{\text{Regional Adjustment (RA)}} \quad \text{with} \quad Y_{it}^H = \underbrace{\frac{Y_{it} + Y_{it-1} + Y_{it-2}}{3}}_{\text{Rebased Benchmark}}$$

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- ▶ **Regional Adjustment $\Rightarrow$ Adverse Selection (AS)** [▶ details](#)
Ex-ante more efficient ACOs can earn SS rewards without changing spending.

Research Questions and Contributions

- **Research questions**
 - ▶ Is there evidence of Ratchet Effect (RE) and Adverse Selection (AS)?
 - ▶ How do RE and AS affect MSSP savings?
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- ▶ Structural quantification of their impact on MSSP savings.
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- **Literature contributions**

- ▶ Financial incentives for healthcare providers.

Gaynor et al. (2004); Ho & Pakes (2014); Eliason et al. (2018); Einav et al. (2021)

- ▶ Unnecessary spending in the U.S.

Fisher et al. (2009); Skinner and Fisher (2010); Cutler (2010); Berwick and Hackbarth (2012); Shrank et al. (2019)

- ▶ MSSP design and its limitations.

McWilliams et al. (2020); Chernew et al. (2023); Aswani et al. (2019); Reddig (2020)

- **ACO Public Use Files (2013 - 2022):**
 - ▶ ACOs financial data: benchmark, spending prior to FFS, spending under MSSP, regional expenditure, beneficiaries demographics.
 - ▶ ACOs Participants list.
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- **“Potential ACOs”** = coalition of providers that resemble the actual ACOs, but do not participate in the MSSP:
 - ▶ AHRQ Compendium of U.S. Health Systems.
 - ▶ Hospital-provider affiliation data from the Physician Compare database.
 - ▶ Physician Shared Patients Patterns data.

- Ratchet Effect: Exploit 2019 benchmark rule change [» details](#)

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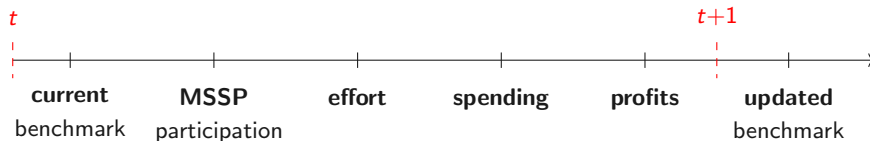
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Structural Model



- (1) Participation [» details](#)

$$d_{it} = 1 \iff V_{it}^{MSSP}(B_{it}) > V_{it}^{FFS}$$

- (3) Spending [» details](#)

$$Y_{it}^{MSSP} = Y_{it}^{FFS} - e_{it}, \quad e_{it} \geq 0$$

$$\ln Y_{it}^{FFS} = \ln Y_{it-1}^{FFS} + \ln \text{tf}_{it} + \ln \text{rr}_{it} + u_{it}$$

- (2) Dynamic effort choice [» Euler eq.](#)

$$e_{it}^* = \underset{e_{it}}{\operatorname{argmax}} \left\{ \pi_{it}^{MSSP}(e_{it}) + \delta \mathbb{E} [V_{it+1}(B_{it+1})] \right\}$$

- (4) Per-period MSSP profits [» details](#)

$$\pi_{it}^{MSSP}(e_{it}) = SS(Y_{it}^{MSSP}, B_{it}) - C(e_{it}) - F_i$$

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Primitive and Estimation

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 - ▶ cost primitives = observed heterogeneity + unobserved heterogeneity

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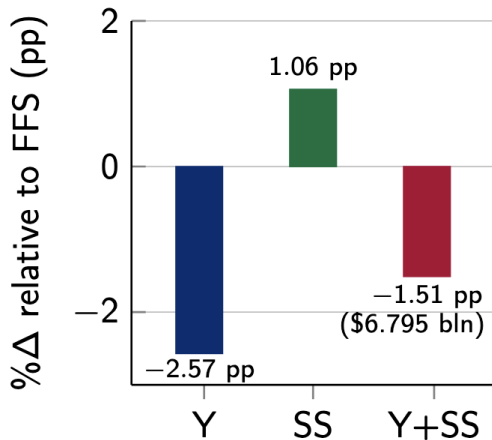
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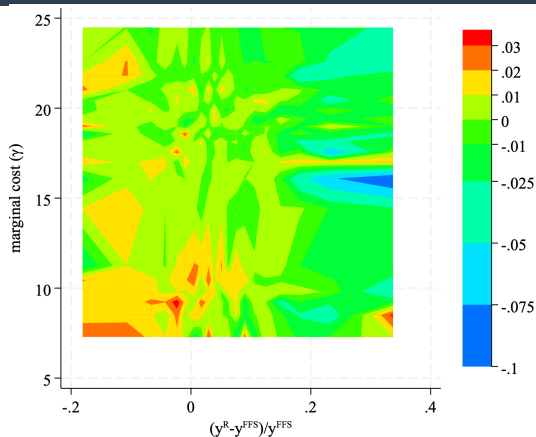
- Likelihood Function. [▶ MLE](#) and EM algorithm (Arcidiacono Miller 2011). [▶ EM](#)

- Estimated Primitives: [▶ effort cost](#) and [▶ fixed costs](#)

Does MSSP generate savings relative to FFS?



Net Spending (Y), Shared Savings (SS),
Total Spending (Y + SS)



x-axis = adverse selection incentive

y-axis = marginal cost of effort

z-axis = $(Y - y^{FFS})/y^{FFS}$

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- ▶ **Alternative design: Conditional Regionalization** [▶▶ Results](#)
 - apply regionalization only when it rewards *real* savings,
 - do not regionalize when rewards come purely from being ex-ante efficient. [▶▶ details](#)

Key Takeaways

- MSSP can be a **win-win** for both healthcare providers (higher profits) and Medicare (lower wasteful spending).
- Issues with current MSSP benchmarking design:
 - ▶ **Ratchet Effect**
 - ▶ **Adverse Selection**and both have a negative impact on MSSP savings.
- There is a **tradeoff** between them RE and AS.
- **Policy Recommendation:** to mitigate the RE while preventing AS, Medicare should apply the regional adjustment to the ACOs the reduce their spending relative to their historical level.

Thank You!

email your comments at cappelae@bc.edu

Part I

Appendix

Type of waste	Berwick & Hackbarth (2011)		Shrank et al. (2019)	
	Billions (2011)	% total (2011)	Billions (2019)	% total (2019) ^b
Care delivery ^a	102–154	3.8–5.7	102–166	2.7–4.4
Care coordination ^a	25–45	0.9–1.7	27–78	0.7–2.1
Overtreatment ^a	158–226	5.9–8.4	76–101	2.0–2.7
Administrative complexity	107–389	4.0–14.4	267	7.0
Pricing failure	84–178	3.1–6.6	231–241	6.1–6.3
Fraud and abuse	82–272	3.0–10.1	59–84	1.6–2.2
Total	558–1,263	20.7–46.8	760–935	20.0–24.6

^a Care delivery, care coordination, and overtreatment are the components of *clinical waste*.

^b 2019 shares are based on projected 2019 national health spending.

^{a, b} Numbers based on Berwick and Hackbarth, 2012; Shrank and colleagues, 2019; and National Health Expenditure Account data extracted from the Peterson-KFF Health System Tracker.

Illustrative Example: Incentives under FFS

Consider a patient with chronic heart failure

- He/she is seen by both a cardiologist and a primary care physician.
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- Provider's cost is \$800, so the margin is \$200.
- Cost of coordination to avoid duplicate test is \$100.

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Providers Incentives:

- The choice is between a \$200 profit and a \$100 loss.
- There is **no financial incentive to avoid the duplicate test.**

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Suppose these providers form an ACO that participates in the MSSP.

- Each provider is still paid with FFS system.
- Under FFS the annual spending for this patient is $B = \$10,000$.
- The duplicate test is avoided, and spending falls to $Y = \$9,000$.
- Medicare saves $B - Y = \$1,000$
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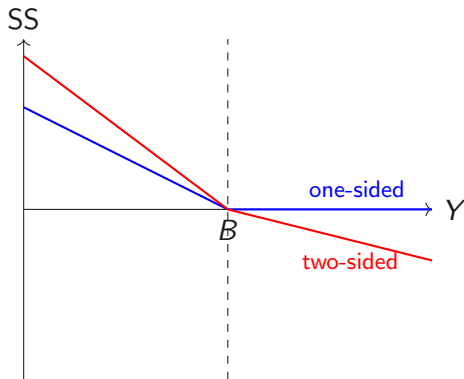
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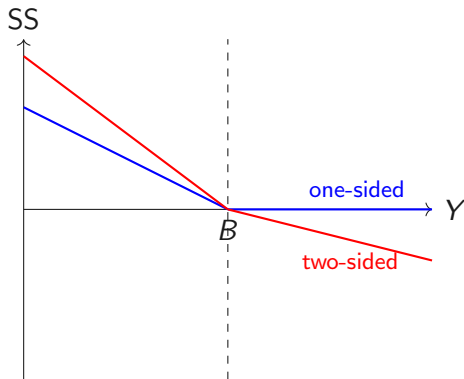
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Shared Savings Contract



- Under one-sided contract:
 - ▶ If $Y < B$, **ACO's reward**
 $SS = 0.5 \cdot (B - Y) > 0$
 - ▶ If $Y \geq B$, **ACO's penalty** $SS = 0$
- Under two-sided contract:
 - ▶ If $Y < B$, **ACO's reward**
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- **Ratchet Effect (RE):** incentives to delay $Y_{it} \downarrow$ to keep high benchmark.

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- **Regionally Adjusted Benchmark:**

Medicare Regional
Average Spending

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$$B_{it+1} = Y_{it}^H + \underbrace{\lambda \left(Y_{it}^R - Y_{it}^H \right)}_{\text{Regional Adjustment (RA)}} \quad \text{with} \quad \lambda \in [0, 1]$$

- ACOs' selection incentive:

The Benchmark Formula and its issues: Adverse Selection

- Regionally Adjusted Benchmark:

Medicare Regional
Average Spending

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- ACOs' selection incentive:

- ▶ If $Y_{it}^{FFS} < Y_{it}^R \Rightarrow RA > 0 \Rightarrow$ Earn SS for free since $B_{it} > Y_{it}^{FFS} \Rightarrow$ **select in (stay)**

The Benchmark Formula and its issues: Adverse Selection

- Regionally Adjusted Benchmark:

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The Benchmark Formula and its issues: Adverse Selection

- Regionally Adjusted Benchmark:

Medicare Regional
Average Spending

$$B_{it+1} = Y_{it}^H + \underbrace{\lambda(Y_{it}^R - Y_{it}^H)}_{\text{Regional Adjustment (RA)}} \quad \text{with} \quad \lambda \in [0, 1]$$

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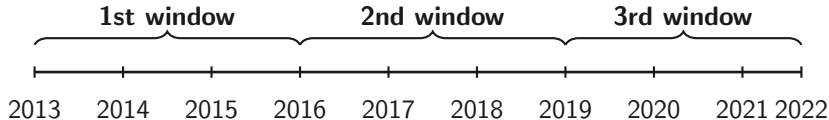
- ▶ If $Y_{it}^{FFS} < Y_{it}^R \Rightarrow RA > 0 \Rightarrow$ Earn SS for free since $B_{it} > Y_{it}^{FFS} \Rightarrow$ **select in (stay)**
- ▶ If $Y_{it}^{FFS} > Y_{it}^R \Rightarrow RA < 0 \Rightarrow$ Do not earn SS even if $Y_{it} < Y_{it-1}^H \Rightarrow$ **select out (exit)**

- Adverse Selection (AS):** ACOs that are ex-ante more efficient than the region ($Y_{it}^{FFS} < Y_{it}^R$) are more likely to join the MSSP since they can earn SS without $Y_{it} \downarrow$.

Rebasement Windows: Pre-2019 vs 2019+

[» back](#)

Pre-2019 rule: 3-year long windows; all years count for rebasement



2019+ rule: 5-year long windows; first 2 years do not count for rebasement

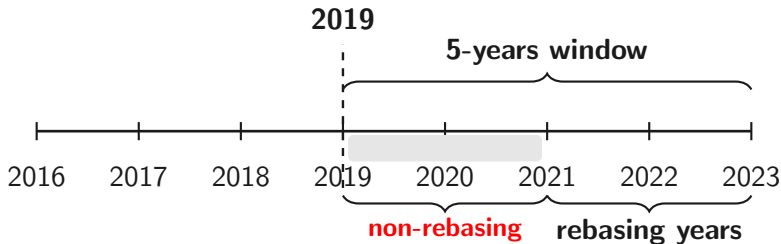


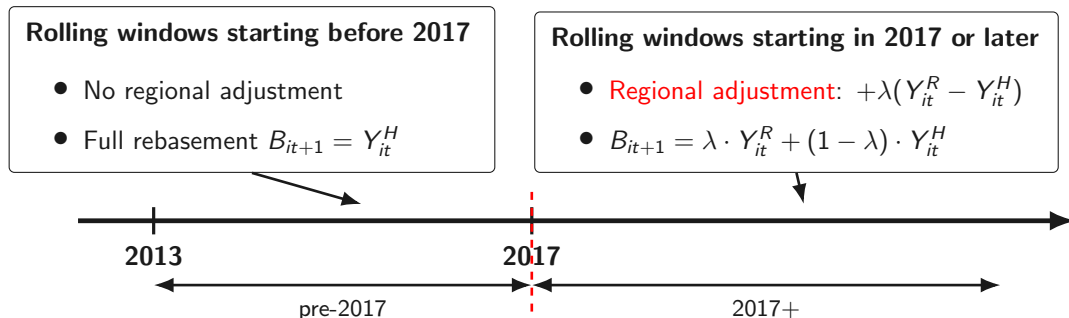
Table 1: Ratchet Effect: Two Way Fixed Effects Regressions

	Full sample	Entry year < 2019
Non-Rebasing Year	0.943*** (0.210)	0.907*** (0.227)
Observations	4,519	4,133
R-squared	0.180	0.186
Mean	2.324	2.235
SD	4.173	4.207

Dependent Variable: CMS Savings Rate = $(B_{it} - Y_{it})/B_{it}$.

Year FE, ACO FE and Agreement Period (AP) are included.

Standard errors are clustered at ACO level.



- **Identification:** Exploit quasi-experimental variation in RA Policy.
 - ▶ $RA_{it} = \mathbb{1}\{\text{RA is applied to ACO } i \text{ in year } t\}$ + staggered rollout across ACO cohorts
 - ▶ $RA_{it}^+ = \mathbb{1}\{B_{it+1} > Y_{it}^H \text{ due to } RA_{it} > 0\}$ + sign depends on pre-policy factors

Table 2: Adverse Selection: Logistic Regressions

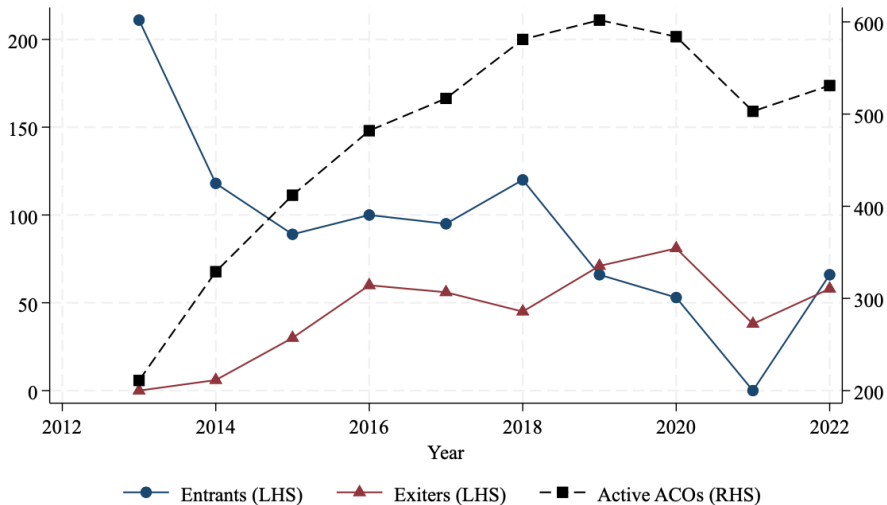
	(1)	(2)	(3)	(4)
Regional Adj	3.641** (1.159)	3.693*** (0.739)	3.735* (2.615)	3.773*** (0.740)
Positive Adj	0.257*** (0.077)	0.266*** (0.036)	0.299** (0.146)	0.296*** (0.037)
McFadden R ²	0.157	0.155	0.141	0.139
AP FE	Yes	No	Yes	No
Year FE	Yes	Yes	No	No

Dependent Variable: Indicator for Exit from MSSP

Coefficients represents the odds ratio of Exit

AP FE stands for agreement period (windows) fixed effects

Participation in the MSSP



Data Source: CMS - MSSP Performance Data 2013-2022

Appendix: Participation condition

- At the beginning of period t , the ACO observes current benchmark B_{it} and decides whether to participate in MSSP.

- Let

$$V_{it}^{MSSP}(B_{it}) = \max_{e_{it} \geq 0} \left\{ \pi_{it}^{MSSP}(e_{it}) + \delta \mathbb{E}[V_{it+1}(B_{it+1})] \right\}$$

denote the value from participation, and let

$$V_{it}^{FFS}$$

denote the value from remaining under FFS.

- Participation decision:

$$d_{it} = 1 \iff V_{it}^{MSSP}(B_{it}) > V_{it}^{FFS}$$

Appendix: Dynamic effort choice

- Conditional on participation, effort solves

$$e_{it}^* = \operatorname{argmax}_{e_{it} \geq 0} \left\{ \pi_{it}^{MSSP}(e_{it}) + \delta \mathbb{E}[V_{it+1}(B_{it+1})] \right\}.$$

- First-order condition:

$$\frac{\partial \pi_{it}^{MSSP}}{\partial e_{it}} + \delta \mathbb{E} \left[\frac{\partial V_{it+1}}{\partial B_{it+1}} \cdot \frac{\partial B_{it+1}}{\partial e_{it}} \right] = 0.$$

- Interpretation:

- ▶ **Marginal benefit today:** more effort lowers current spending and raises current shared savings.
- ▶ **Marginal loss tomorrow:** more effort may lower future benchmark B_{it+1} through rebasement, reducing future rents.

- This is the source of the **ratchet effect**: current cost reduction can worsen future incentives. [◀ back](#)

Appendix: Spending process

- MSSP spending equals counterfactual FFS spending minus effort:

$$Y_{it}^{MSSP} = Y_{it}^{FFS} - e_{it}, \quad e_{it} \geq 0.$$

- Counterfactual FFS spending evolves as

$$\ln Y_{it}^{FFS} = \ln Y_{it-1}^{FFS} + \ln \text{tf}_{it} + \ln \text{rr}_{it} + u_{it},$$

where:

- ▶ tf_{it} is the inflation / utilization trend factor,
 - ▶ rr_{it} is the risk ratio,
 - ▶ $u_{it} \sim N(0, \sigma_i^2)$ is an idiosyncratic spending shock.
- Interpretation:
 - ▶ Y_{it}^{FFS} is the spending level absent MSSP effort.
 - ▶ Effort shifts actual MSSP spending below this counterfactual.

Appendix: Per-period MSSP profits

- Per-period profits under MSSP are

$$\pi_{it}^{MSSP}(e_{it}) = SS(Y_{it}^{MSSP}, B_{it}) - C(e_{it}) - F_i.$$

- Shared savings are generated when spending falls below benchmark:

$$SS(Y_{it}^{MSSP}, B_{it}) = s_i \cdot \max\{B_{it} - Y_{it}^{MSSP}, 0\}.$$

- Variable effort cost:

$$C(e_{it}) = \frac{1}{2} \gamma_i \frac{e_{it}^2}{Y_{it}^{FFS}},$$

where $\gamma_i > 0$ governs the curvature of the cost of cost-reduction effort.

- F_i are fixed costs of participation in MSSP:

Likelihood Function

- Let $\theta = (\beta_C, \alpha_C, \beta_F, \alpha_F, \rho)$.

Likelihood Function

- Let $\theta = (\beta_C, \alpha_C, \beta_F, \alpha_F, \rho)$.
- I observe $\{(d_{it}, y_{it})\}_{t=1}^{T_i}$ and $\{\Omega_{it}\}_{t=1}^{T_i}$.

Likelihood Function

- Let $\theta = (\beta_C, \alpha_C, \beta_F, \alpha_F, \rho)$.
- I observe $\{(d_{it}, y_{it})\}_{t=1}^{T_i}$ and $\{\Omega_{it}\}_{t=1}^{T_i}$.
- The likelihood function for ACO i :

$$L_i(\theta) = \sum_k q_{ik} \left[\prod_{t=1}^{T_i-1} \phi(y_{it}, \Omega_{it}, k; \theta) \Pr(d_{it} = 1 | \Omega_{it}, k; \theta) \right] \times \Pr(d_{iT_i} = 0 | \Omega_{iT_i}, k; \theta)$$

where q_{ik} s are the unobserved type probabilities , and

$$\phi(y_{it} | \Omega_{it}, k) = \frac{1}{\sqrt{2\pi\sigma_{it}^2}} \exp\left(-\frac{(y_{it} - \mu_{it} - e_{it})^2}{2\sigma_{it}^2}\right), \quad \mu_{it} = y_{it-1}^{FFS} \cdot \eta_{it} \cdot rr_{it}$$

and

$$\Pr(d_{it} = 1 | \Omega_{it}, k; \theta) = \frac{1}{1 + \exp(v_{it}^F - v_{it}^M)}$$

» back

Likelihood Function

- Let $\theta = (\beta_C, \alpha_C, \beta_F, \alpha_F, \rho)$.
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and

$$\Pr(d_{it} = 1 | \Omega_{it}, k; \theta) = \frac{1}{1 + \exp(v_{it}^F - v_{it}^M)}$$

» back

- I estimate θ with an EM algorithm (Arcidiacono Miller 2011).

Expectation Maximization

- Expectation Step:
 - ▶ Use Bayes' Rule to update q_{ik}
 - ▶ Update the population probability of each type, w_k
 - ▶ Using $\theta^{(m)}$ and $p^{(m)}$, update the CCP

$$p_1^{(m+1)} = \frac{1}{1 + \exp \left[v_{it}^F(\Omega_{it}, k; p^{(m)}, \theta^{(m)}) - v_{it}^M(\Omega_{it}, k; p^{(m)}, \theta^{(m)}) \right]}$$

Expectation Maximization

- Expectation Step:
 - ▶ Use Bayes' Rule to update q_{ik}
 - ▶ Update the population probability of each type, w_k
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$$p_1^{(m+1)} = \frac{1}{1 + \exp [v_{it}^F(\Omega_{it}, k; p^{(m)}, \theta^{(m)}) - v_{it}^M(\Omega_{it}, k; p^{(m)}, \theta^{(m)})]}$$

- Maximization Step: ▶ simulated value function

$$\theta^{(m+1)} = \operatorname{argmax}_{\theta} \sum_i \sum_k q_{ik} \log L_{ik}(\theta | q_{ik}^{(m+1)}, p_1^{(m+1)}, \Omega_{it}, k))$$

Simulated Value Function

To evaluate v_{it}^M , we use forward simulation to simulate S paths of participation effort, spending and profits

$$v_{it}^M \approx \frac{1}{S} \sum_{s=1}^S \sum_{h=0}^H \delta^h \pi_{i,t+h}^{(s)}$$

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To evaluate v_{it}^M , we use forward simulation to simulate S paths of participation effort, spending and profits

$$v_{it}^M \approx \frac{1}{S} \sum_{s=1}^S \sum_{h=0}^H \delta^h \pi_{i,t+h}^{(s)}$$

Given some value of the parameters θ , each simulated path uses:

- d_{it} draws from Kernel probabilities $\hat{Pr}(d_{it} = 1|\Omega_{it})$
- e_{it} from the Euler equation [▶ euler equation](#)
- y_{it} from the spending equation
- $\tilde{y}_{it}$ and b_{it} move forward according to the MSSP rules. [▶ back](#)

Counterfactual: Turning off RE

- Prevent Y_{it} from affecting B_{it+1}

$$B_{it+1} = Y_{it}^H + \lambda(Y_{it}^R - Y_{it}^H) \quad \text{where} \quad Y_{it}^H = \frac{Y_{it}^{MSSP} + Y_{it-1}^{MSSP} + Y_{it-2}^{MSSP}}{3}$$

↑
set $\lambda = 1$
to turn off RE

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- Y_{it} no longer affects $B_{it+1} \Rightarrow$ **No more RE**

Counterfactual: Turning off RE

- Prevent Y_{it} from affecting B_{it+1}

$$B_{it+1} = Y_{it}^H + \lambda(Y_{it}^R - Y_{it}^H) \quad \text{where} \quad Y_{it}^H = \frac{Y_{it}^{MSSP} + Y_{it-1}^{MSSP} + Y_{it-2}^{MSSP}}{3}$$

↑
set $\lambda = 1$
to turn off RE

- Y_{it} no longer affects $B_{it+1} \Rightarrow$ **No more RE**
- Full Regional Benchmark**

$$B_{it+1} = Y_{it}^R$$

$\Rightarrow$ AS as is at its strongest.

Estimation Results: Variable Costs

	K=1	K=2	K=3	K=4
Risk Score	0.541** (0.156)	0.493*** (0.137)	0.435* (0.053)	0.451*** (0.061)
Beneficiaries	-0.241*** (0.059)	-0.193*** (0.039)	0.135* (0.035)	0.173*** (0.040)
Hospital	1.641** (0.316)	1.523** (0.358)	1.735** (0.415)	1.613** (0.421)
Type 1		0.302*** (0.045)		
Type 2		0.217*** (0.052)	0.134** (0.043)	
Type 3		0.147** (0.042)	0.094** (0.028)	0.056 (0.053)
GOF	0.852	0.881	0.936	0.904

(i) Coefficients = $\Delta\$$ in the marginal cost of reducing spending by 1%.

(ii) Standard errors computed via bootstrapping, with 1000 resampling.

Estimation Results: Fixed Costs

	K=1	K=2	K=3	K=4
Beneficiaries	0.142** (0.059)	0.194** (0.039)	0.136** (0.042)	0.154** (0.041)
Hospital	0.640** (0.116)	0.554** (0.158)	0.637** (0.135)	0.643** (0.121)
Type 1		0.252*** (0.045)		
Type 2		0.253*** (0.052)	0.216*** (0.043)	
Type 3		0.247** (0.068)	0.214** (0.057)	0.156* (0.083)
GOF	0.852	0.881	0.936	0.904

- (i) Coefficients = percentage change in fixed cost for a unit increase in x .
(ii) Standard errors computed via bootstrapping, with 1000 resampling.

Counterfactual: Turning off AS

- Remove the Regional Adjustment

$$B_{it+1} = Y_{it}^H + \lambda(Y_{it}^R - Y_{it}^H)$$



set $\lambda = 0$
to turn off AS

Counterfactual: Turning off AS

- Remove the Regional Adjustment

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- RA does not enter $B_{it+1} \Rightarrow$ **No more AS**

Counterfactual: Turning off AS

- Remove the Regional Adjustment

$$B_{it+1} = Y_{it}^H + \lambda(Y_{it}^R - Y_{it}^H)$$

↑
set $\lambda = 0$
to turn off AS

- RA does not enter $B_{it+1} \Rightarrow$ **No more AS**
- Fully Rebased Benchmark**

$$B_{it+1} = Y_{it}^H$$

$\Rightarrow$ RE as is at its strongest.

New Benchmark Design: Conditional Regionalization

- Status quo benchmark: $B_{it+1} = \underbrace{Y_{it}^H}_{\text{rebased benchmark}} + \underbrace{\lambda \cdot (Y_{it}^R - Y_{it}^H)}_{\text{regional adjustment}}$

Issue: RA is applied independently from the ACOs' spending behavior during the MSSP.

New Benchmark Design: Conditional Regionalization

- Status quo benchmark: $B_{it+1} = \underbrace{Y_{it}^H}_{\text{rebased benchmark}} + \underbrace{\lambda \cdot (Y_{it}^R - Y_{it}^H)}_{\text{regional adjustment}}$

Issue: RA is applied independently from the ACOs' spending behavior during the MSSP.

- Conditional Regionalization:**

Historical vs. Regional spending	Current vs Historical spending	
	Saving: $Y_{it} < Y_{it}^H$	Loss: $Y_{it} \geq Y_{it}^H$
More efficient $Y_{it}^H < Y_{it}^R$	Regionalize $B_{it+1} = Y_{it}^H + \lambda(Y_{it}^R - Y_{it}^H)$	No regionalization $B_{it+1} = Y_{it}^H$
Less efficient $Y_{it}^H \geq Y_{it}^R$	No regionalization $B_{it+1} = Y_{it}^H$	Regionalize $B_{it+1} = Y_{it}^H + \lambda(Y_{it}^R - Y_{it}^H)$

Case 1: More efficient than the region + real savings $\Rightarrow$ **RA is applied.**

New Benchmark Design: Conditional Regionalization

- Status quo benchmark: $B_{it+1} = \underbrace{Y_{it}^H}_{\text{rebased benchmark}} + \underbrace{\lambda \cdot (Y_{it}^R - Y_{it}^H)}_{\text{regional adjustment}}$

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Case 2: More efficient than the region + no real savings $\Rightarrow$ **RA is not applied.**

New Benchmark Design: Conditional Regionalization

- Status quo benchmark: $B_{it+1} = \underbrace{Y_{it}^H}_{\text{rebased benchmark}} + \underbrace{\lambda \cdot (Y_{it}^R - Y_{it}^H)}_{\text{regional adjustment}}$

Issue: RA is applied independently from the ACOs' spending behavior during the MSSP.

- Conditional Regionalization:**

Historical vs. Regional spending	Current vs Historical spending	
	Saving: $Y_{it} < Y_{it}^H$	Loss: $Y_{it} \geq Y_{it}^H$
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Less efficient $Y_{it}^H \geq Y_{it}^R$	No regionalization $B_{it+1} = Y_{it}^H$	Regionalize $B_{it+1} = Y_{it}^H + \lambda(Y_{it}^R - Y_{it}^H)$

Case 3: Less efficient than the region + real savings $\Rightarrow$ **RA is not applied.**

New Benchmark Design: Conditional Regionalization

- Status quo benchmark: $B_{it+1} = \underbrace{Y_{it}^H}_{\text{rebased benchmark}} + \underbrace{\lambda \cdot (Y_{it}^R - Y_{it}^H)}_{\text{regional adjustment}}$

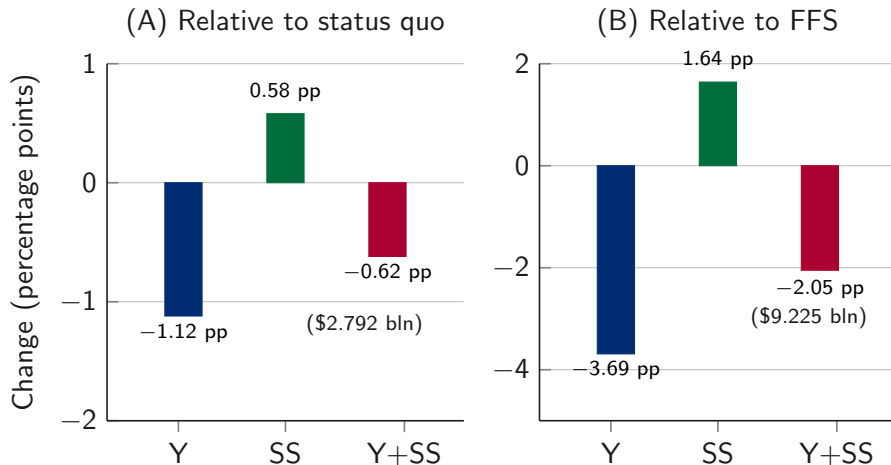
Issue: RA is applied independently from the ACOs' spending behavior during the MSSP.

- Conditional Regionalization:**

Historical vs. Regional spending	Current vs Historical spending	
	Saving: $Y_{it} < Y_{it}^H$	Loss: $Y_{it} \geq Y_{it}^H$
More efficient $Y_{it}^H < Y_{it}^R$	Regionalize $B_{it+1} = Y_{it}^H + \lambda(Y_{it}^R - Y_{it}^H)$	No regionalization $B_{it+1} = Y_{it}^H$
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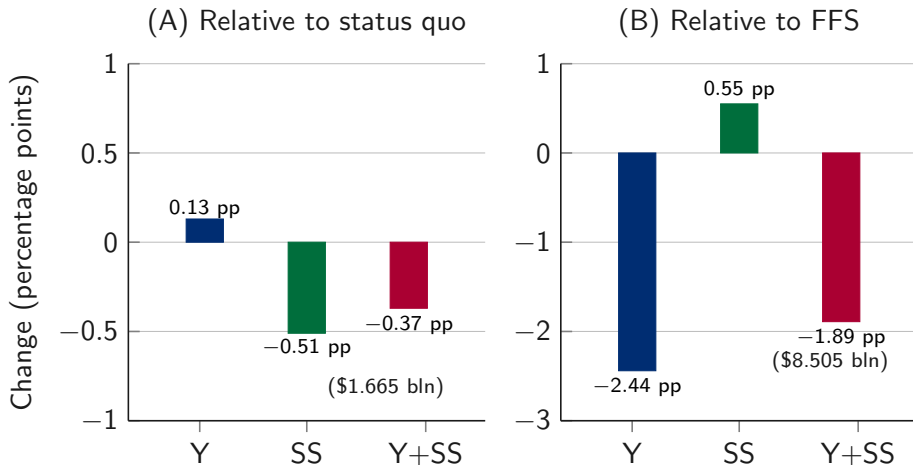
Case 4: Less efficient than the region + no real savings $\Rightarrow$ **RA is applied.**

Scenario w/out RE vs. Status Quo and FFS



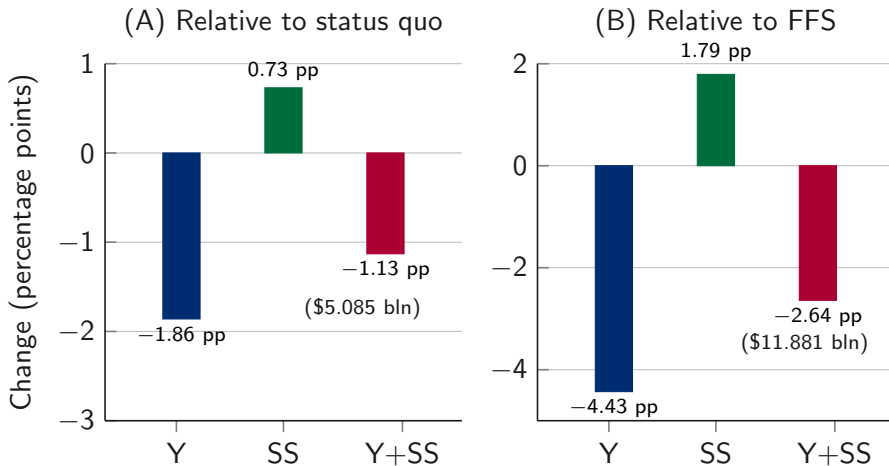
Net Spending (Y), Shared Savings (SS), Total Spending (Y + SS). [▶ back](#)

Scenario w/out AS vs. Status Quo and FFS



Net Spending (Y), Shared Savings (SS), Total Spending (Y + SS). [» back](#)

Conditional Regionalization vs. status quo and FFS



Net Spending (Y), Shared Savings (SS), Total Spending (Y + SS). [» back](#)